

Solução do Problema D6CUBE - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: D6CUBE
- Número de Variáveis: 6184
- Número de Restrições: 415
- Inviabilidade Primal: 2.703e-13
- Inviabilidade Dual: 1.288e-14
- Valor Primal: 3.155e+02
- Valor Dual: 3.103e+02
- Gap: 8.553e-19
- Número de Iterações: 30

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema D6CUBE

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
94	2.300e+00	1	4.441e-16
95	7.717e+00	2	1.332e-14
96	8.500e-01	3	1.667e-01
97	3.250e+01	4	2.500e-01
101	1.000e+00	6	5.000e-01
415	1.700e+00	7	2.981e-01
480	4.848e-14	8	6.013e-01
652	8.615e-14	9	3.333e-01
753	3.247e+01	10	1.364e-01
755	5.000e-01	11	3.525e-02
760	5.533e+00	12	5.595e-01
764	3.333e-02	13	4.811e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
765	2.021e-13	14	5.644e-01
777	1.000e+00	15	5.515e-01
801	3.819e-14	16	7.207e-02
851	1.000e+00	17	6.621e-01
917	2.871e-14	18	9.839e-01
1502	2.000e+00	19	1.045e+00
1511	2.610e+01	20	9.156e-01
1518	2.000e+00	21	6.219e-01
1557	8.615e-14	22	9.810e-01
1686	1.000e+00	23	6.219e-01
1846	2.000e+00	24	6.268e-01
1849	3.100e+00	25	9.156e-01
1906	9.011e-14	26	6.766e-01
1986	3.276e-15	27	1.970e-01
2026	2.300e+00	28	7.363e-01
2027	1.436e-13	29	5.821e-01
2031	6.738e-14	30	9.555e-01
2088	3.109e-15	31	6.268e-01
2176	1.000e+00	32	8.710e-01
2451	1.066e-14	33	3.333e-01
2734	2.550e+00	34	6.735e-01
2746	2.550e+00	35	1.000e+00
3066	1.625e+00	36	9.156e-01
3068	2.212e-13	37	1.045e+00
3072	1.625e+00	38	1.000e+00
3136	5.270e-14	39	1.052e+00
3156	1.158e+00	40	9.156e-01
3158	1.667e-02	41	5.515e-01
3159	1.300e+00	42	9.156e-01
3167	2.608e+00	43	2.372e-01
3175	1.667e-02	44	4.190e-01
3178	7.167e-01	45	5.373e-01
3179	1.108e+00	46	6.766e-01
3212	8.615e-14	47	9.953e-01
3240	2.083e-01	48	4.527e-01
3253	5.033e+00	49	1.002e+00
3261	2.393e+01	50	5.821e-01
3263	1.437e-13	51	9.156e-01
3276	6.167e+00	52	9.761e-01
3277	1.437e-13	53	9.810e-01
3279	2.250e+00	54	5.486e-01
3283	1.350e-13	55	9.555e-01
3288	1.612e+01	56	9.156e-01
3292	9.746e-15	57	9.953e-01
3295	9.746e-15	58	5.373e-01
3304	2.000e+00	59	1.000e+00
3306	5.000e-01	60	4.975e-01
3311	3.333e-02	61	8.310e-01
3347	3.333e-02	62	9.270e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
3481	3.782e-14	63	1.000e+00
3755	1.200e+00	64	1.413e+00
3772	6.083e-01	65	9.156e-01
3773	8.615e-14	66	1.413e+00
3805	4.219e-15	67	1.413e+00
3815	1.015e+01	68	9.156e-01
3915	2.433e+00	69	2.132e-01
3950	3.105e-14	70	8.262e-01
3979	2.755e-14	71	9.156e-01
3992	8.615e-14	72	1.413e+00
4007	8.083e-01	73	9.156e-01
4053	3.536e-15	74	5.870e-01
4055	3.797e-15	75	9.555e-01
4097	8.615e-14	76	9.555e-01
4125	2.998e-15	77	1.014e+00
4126	8.615e-14	78	9.555e-01
4136	3.000e-01	79	1.413e+00
4152	5.945e-14	80	9.555e-01
4155	1.000e+00	81	8.003e-01
4158	9.000e-01	82	1.413e+00
4170	3.333e-02	83	1.000e+00
4176	9.000e-01	84	9.810e-01
4177	3.271e-14	85	9.156e-01
4187	2.000e+00	86	4.975e-01
4190	2.000e+00	87	1.000e+00
4198	1.194e-13	88	1.413e+00
4202	1.000e+00	89	9.156e-01
4261	1.965e-14	90	1.413e+00
4269	8.703e-14	91	1.413e+00
4394	2.354e-14	92	2.675e-01
4399	2.354e-14	93	3.333e-01
4415	3.000e-01	98	1.667e-01
4436	3.000e-01	99	1.667e-01
4437	3.276e-15	100	6.661e-16
4444	6.497e-15	102	2.333e-01
4447	1.739e-14	103	1.685e-01
4450	1.717e+00	104	1.665e-15
4453	6.000e-01	105	5.000e-01
4454	2.583e+00	106	6.169e-01
4457	2.021e-13	107	2.220e-15
4469	2.108e+00	108	5.106e-02
4472	5.592e+00	109	3.386e-01
4476	1.963e-13	111	1.780e-01
4489	6.092e+00	112	6.659e-01
4492	3.819e-14	113	4.104e-01
4495	1.189e-13	114	7.838e-01
4508	7.017e+00	115	3.351e-01
4536	1.250e+00	116	1.685e-01
4537	1.667e-02	117	1.263e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
4548	2.948e-14	118	3.194e-01
4553	5.167e-01	119	1.198e+00
4555	1.667e-02	120	1.695e+00
4578	8.000e-01	121	1.975e-01
4616	1.917e-01	122	3.415e-01
4653	1.667e-02	123	8.044e-01
4657	9.167e-02	124	6.969e-01
4698	1.139e-14	125	7.232e-01
4705	4.848e-14	126	1.246e+00
4713	9.667e-01	127	4.109e-01
4716	1.667e-02	128	7.184e-01
4733	2.871e-14	129	1.216e+00
4822	1.436e-14	130	5.290e-02
4854	4.000e+00	131	6.062e-01
4873	1.350e+01	132	5.718e-01
4877	2.000e+00	133	5.408e-01
4893	6.000e+00	134	1.326e-02
4910	2.000e+00	135	6.124e-01
4917	1.000e+00	136	3.546e-01
4920	1.000e+00	137	6.314e-01
4946	9.746e-15	138	2.056e-01
4974	2.000e+00	139	5.470e-01
5100	3.142e-14	140	5.095e-01
5115	5.167e+00	141	3.333e-01
5138	6.083e-01	142	3.333e-01
5142	6.250e-01	143	5.937e-01
5207	5.000e-01	144	1.216e+00
5212	1.984e-14	145	8.079e-01
5215	7.667e-01	146	6.878e-01
5216	3.775e-15	147	1.633e-01
5308	2.000e+00	148	4.671e-01
5321	5.500e-01	149	1.693e-01
5422	2.667e-01	150	4.695e-01
5443	4.833e-01	151	7.328e-01
5455	4.308e-14	152	7.328e-01
5482	3.000e-01	153	8.030e-01
5492	5.000e-01	154	8.478e-01
5515	4.020e-15	155	5.900e-01
5543	3.333e-02	156	1.216e+00
5673	1.433e+00	157	7.328e-01
5682	1.900e+00	158	3.849e-01
5699	5.017e+00	159	8.478e-01
5717	9.827e-15	160	2.157e-01
5719	9.827e-15	161	8.030e-01
5793	5.000e-01	162	6.034e-01
5800	5.000e-01	163	6.506e-01
5827	1.000e+00	164	1.019e+00
5838	3.109e-15	165	7.665e-01
5854	1.000e+00	166	7.184e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
5887	2.000e+00	167	7.184e-01
5899	1.000e+00	168	1.216e+00
5954	1.948e-14	169	9.106e-01
5955	7.550e-15	170	4.624e-01
5966	7.500e-01	171	8.665e-01
5992	1.483e+00	172	1.693e-01
6098	6.552e-15	173	1.693e-01
6099	8.333e-03	174	3.003e-01
6102	2.083e-01	175	7.184e-01
6120	1.667e-02	176	3.725e-01
6128	9.333e-01	177	5.212e-01
6134	2.667e-01	178	1.216e+00
6138	2.167e-01	179	8.738e-01
6152	3.000e-01	180	3.003e-01
6157	4.833e-01	181	5.000e-01
6170	1.000e+00	182	3.896e-01
6184	1.000e+00	183	7.429e-01
		184	1.072e+00
		185	4.048e-01
		186	7.384e-01
		187	7.286e-01
		188	8.077e-01
		189	7.384e-01
		192	5.000e-01
		195	3.030e-01
		197	4.426e-01
		198	6.219e-01
		199	5.323e-01
		200	1.490e-01
		201	8.424e-02
		202	6.092e-01
		203	3.333e-01
		204	1.306e-02
		205	4.669e-01
		206	3.917e-02
		207	6.950e-01
		208	2.666e-01
		209	6.761e-01
		210	2.318e-01
		211	8.518e-01
		212	1.103e+00
		213	3.003e-01
		214	4.692e-01
		215	5.279e-01
		216	2.608e-01
		217	2.083e-01
		218	3.436e-01
		219	4.323e-01
		220	6.447e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		222	6.777e-01
		223	5.000e-01
		224	4.636e-01
		225	5.068e-01
		226	1.246e+00
		227	5.353e-01
		228	9.674e-01
		229	4.633e-01
		230	6.241e-01
		231	2.033e-01
		232	4.247e-01
		233	7.374e-01
		234	4.333e-01
		235	6.115e-01
		236	6.178e-01
		237	1.002e+00
		238	6.447e-01
		239	4.250e-01
		240	6.895e-01
		241	4.636e-01
		242	8.402e-01
		243	1.092e+00
		244	6.241e-01
		245	4.505e-01
		246	8.402e-01
		247	6.366e-01
		248	2.999e-01
		249	6.430e-01
		250	9.803e-01
		251	4.302e-01
		252	1.013e+00
		253	1.018e+00
		254	4.727e-01
		255	9.422e-01
		256	8.784e-01
		257	6.429e-01
		258	8.774e-01
		259	7.326e-01
		260	1.087e+00
		261	4.279e-01
		262	9.248e-01
		263	1.169e+00
		264	6.831e-01
		265	7.853e-01
		266	5.870e-01
		267	7.326e-01
		268	5.511e-01
		269	1.988e-01
		270	5.870e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		271	1.089e+00
		272	3.092e-01
		273	7.023e-01
		274	6.125e-01
		275	3.594e-01
		276	8.402e-01
		277	6.766e-01
		278	6.625e-01
		279	4.490e-01
		280	8.402e-01
		281	1.031e+00
		282	1.338e+00
		283	8.920e-01
		284	9.911e-01
		285	1.338e+00
		286	9.696e-01
		287	4.873e-01
		288	9.555e-01
		289	9.139e-01
		290	6.140e-01
		291	1.092e+00
		292	9.156e-01
		293	5.821e-01
		294	5.921e-01
		295	7.520e-01
		296	8.402e-01
		297	7.023e-01
		298	6.957e-01
		299	8.402e-01
		300	8.484e-01
		301	6.777e-01
		302	5.638e-01
		303	7.146e-01
		304	8.993e-01
		305	2.742e-01
		306	9.502e-01
		307	1.076e+00
		308	1.120e+00
		309	1.076e+00
		310	8.608e-01
		311	7.998e-01
		312	8.005e-01
		313	1.001e+00
		314	3.963e-01
		315	6.625e-01
		316	7.520e-01
		317	5.245e-01
		318	8.402e-01
		319	1.177e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		320	6.766e-01
		321	7.023e-01
		322	4.235e-01
		323	8.402e-01
		324	9.156e-01
		325	9.911e-01
		326	8.402e-01
		327	1.338e+00
		328	1.338e+00
		329	9.911e-01
		330	9.696e-01
		331	7.472e-01
		332	7.184e-01
		333	8.402e-01
		334	9.911e-01
		335	9.156e-01
		336	2.981e-01
		337	7.184e-01
		338	1.685e-01
		339	6.011e-01
		340	1.025e+00
		341	9.156e-01
		342	9.868e-01
		343	7.152e-01
		344	1.090e+00
		345	1.120e+00
		346	7.421e-01
		347	4.633e-01
		348	6.931e-01
		349	8.800e-01
		350	8.402e-01
		351	4.427e-01
		352	1.076e+00
		353	7.648e-01
		354	6.218e-01
		355	9.156e-01
		356	9.156e-01
		357	8.402e-01
		358	1.076e+00
		359	9.262e-01
		360	1.132e+00
		361	9.156e-01
		362	1.338e+00
		363	1.338e+00
		364	8.115e-01
		365	5.591e-01
		366	1.104e+00
		367	1.489e+00
		368	1.120e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		369	8.402e-01
		370	1.031e+00
		371	1.338e+00
		372	9.911e-01
		373	1.263e+00
		374	1.338e+00
		375	9.056e-01
		376	9.911e-01
		377	1.045e+00
		378	1.031e+00
		379	9.911e-01
		380	9.646e-01
		381	1.338e+00
		382	1.338e+00
		383	1.141e+00
		384	1.141e+00
		385	7.939e-01
		386	7.724e-01
		387	4.975e-01
		388	9.156e-01
		389	3.849e-01
		390	6.591e-01
		391	7.184e-01
		392	1.031e+00
		393	5.472e-01
		394	9.248e-01
		395	6.895e-01
		396	9.911e-01
		397	8.402e-01
		398	4.075e-01
		399	9.156e-01
		400	7.697e-01
		401	4.699e-01
		402	9.187e-01
		403	1.028e+00
		404	7.587e-01
		405	8.789e-01
		406	3.464e-01
		407	5.167e-01
		408	8.003e-01
		409	5.735e-01
		410	6.421e-02
		411	6.413e-01
		412	3.333e-01
		413	7.089e-01
		414	3.831e-01
		416	5.547e-01
		417	7.296e-01
		418	6.990e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		419	3.587e-01
		420	2.401e-01
		421	7.692e-01
		422	7.599e-01
		423	8.850e-01
		424	4.592e-01
		425	6.209e-01
		426	1.132e+00
		427	7.033e-01
		428	9.113e-01
		429	6.852e-01
		430	4.889e-01
		431	3.148e-01
		432	1.079e+00
		433	9.967e-01
		434	9.213e-01
		435	9.213e-01
		436	6.335e-01
		437	5.488e-01
		438	5.543e-01
		439	1.038e+00
		440	8.893e-01
		441	3.638e-01
		442	7.089e-01
		443	1.245e+00
		444	2.853e-01
		445	1.200e+00
		446	9.896e-01
		447	9.213e-01
		448	5.857e-01
		449	4.889e-01
		450	1.005e+00
		451	6.820e-01
		452	4.600e-01
		453	6.196e-01
		454	6.508e-01
		455	2.013e-01
		456	9.213e-01
		457	1.090e+00
		458	9.113e-01
		459	8.293e-01
		460	8.910e-01
		461	1.069e+00
		462	8.409e-01
		463	9.213e-01
		464	3.711e-01
		465	1.825e+00
		466	1.879e+00
		467	9.959e-01
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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
		468	8.011e-01
		469	9.213e-01
		470	8.313e-01
		471	1.354e+00
		472	9.522e-01
		473	1.354e+00
		474	1.836e+00
		475	1.278e+00
		476	3.757e-01
		477	5.273e-01
		478	1.051e+00
		479	7.677e-01
		481	6.341e-01
		482	1.076e+00
		483	6.793e-01
		484	9.764e-01
		485	5.415e-01
		486	1.051e+00
		487	6.169e-01
		488	1.298e-01
		489	6.547e-01
		490	9.068e-01
		491	8.999e-01
		492	5.671e-01
		493	5.981e-01
		494	1.205e+00
		495	1.049e+00
		496	4.679e-01
		497	1.150e+00
		498	5.176e-01
		500	7.589e-01
		501	4.281e-01
		502	3.587e-01
		503	2.318e-01
		504	1.080e+00
		505	3.164e-01
		506	9.363e-01
		507	9.405e-01
		508	9.099e-01
		509	7.545e-01
		510	9.405e-01
		511	9.256e-01
		512	2.953e-01
		513	7.787e-01
		514	4.245e-01
		515	6.326e-01
		516	5.752e-01

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema D6CUBE

Coordenada y	Valor y (Dual)
2	2.099e-02
4	1.254e-01
14	3.469e-01
16	4.985e-01
17	5.444e-02
42	2.298e-01
43	9.730e-02
44	5.132e-01
45	5.127e-01
47	4.113e-01
53	1.225e-01
74	4.284e-01
77	2.782e-01
79	8.241e-02
81	5.380e-03
87	6.431e-02
88	5.810e-02
94	1.508e-01
100	8.611e-02
102	2.445e-02
103	2.236e-03
105	2.907e-01
109	1.887e-01
114	4.934e-03
116	1.157e-01
125	4.198e-01
126	6.230e-02
128	4.332e-01
129	2.617e-01
130	2.299e-01
133	1.142e-02
135	3.928e-02
136	1.951e-01
137	5.194e-02
138	3.281e-01
139	3.074e-02
141	2.744e-02
142	1.075e-01
143	7.192e-02
144	1.088e-01
145	4.669e-02
146	1.107e-02
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Coordenada y	Valor y (Dual)
147	4.339e-02
148	9.551e-04
150	1.577e-01
152	4.773e-02
153	1.825e-01
154	1.584e-02
155	9.442e-02
156	7.209e-02
157	2.967e-01
159	2.016e-01
161	1.389e-01
163	6.229e-01
164	5.954e-01
165	2.051e-01
167	2.877e-02
168	6.398e-02
172	1.430e-01
173	1.123e-01
174	9.439e-02
175	6.369e-02
178	5.920e-02
180	3.357e-01
188	2.157e-01
189	7.057e-01
195	9.353e-02
196	2.765e-01
198	5.409e-02
199	9.506e-02
200	2.634e-01
205	5.830e-02
206	6.276e-02
208	8.995e-02
213	2.983e-01
214	4.784e-01
215	2.166e-01
216	2.448e-01
217	7.596e-01
219	4.279e-01
220	1.738e-02
221	7.235e-02
227	3.372e-01
229	2.204e-01
230	5.036e-02
233	6.907e-01
234	9.245e-01
239	7.016e-01
240	4.762e-01
241	7.913e-01
242	1.401e-01
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Coordenada y	Valor y (Dual)
243	1.450e-01
244	1.319e-01
245	2.200e-01
246	7.273e-02
248	5.737e-02
249	1.455e-01
251	6.699e-01
254	2.364e-02
257	4.287e-01
258	1.354e-01
259	2.478e-01
260	2.308e-02
261	2.291e-01
262	8.533e-02
263	1.507e-01
265	2.217e-01
266	8.780e-02
267	8.519e-02
268	3.309e-01
271	5.386e-01
272	8.315e-02
274	2.246e-01
275	2.356e-02
276	1.631e-01
277	2.847e-01
278	2.507e-01
280	2.554e-02
282	2.758e-02
284	2.402e-01
288	2.290e-01
289	5.732e-01
290	7.880e-02
291	3.314e-01
292	3.124e-01
293	5.578e-01
294	5.194e-01
295	1.540e-01
296	1.271e-01
297	1.712e-01
298	4.230e-02
299	1.174e-01
300	2.867e-01
301	1.443e-01
302	4.687e-01
303	9.536e-02
306	2.901e-01
307	1.099e-01
308	2.638e-01
310	6.484e-02
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Coordenada y	Valor y (Dual)
314	6.867e-02
317	9.451e-03
319	1.488e-01
320	3.580e-01
321	4.138e-01
322	3.275e-01
323	4.668e-01
324	3.805e-01
325	3.958e-01
326	5.306e-01
327	1.833e-01
328	1.915e-02
329	1.092e-01
332	2.274e-01
333	1.468e-02
336	1.620e-01
337	3.987e-01
338	2.242e-01
339	2.392e-01
340	4.235e-01
344	2.922e-01
346	2.937e-02
347	9.420e-03
348	3.618e-01
357	2.041e-01
358	4.613e-01
359	5.344e-01
360	2.274e-01
361	5.093e-01
364	4.366e-01
365	2.714e-01
366	7.874e-01
367	1.020e+00
368	2.535e-01
369	1.508e-01
372	7.073e-03
373	5.838e-01
376	8.107e-02
377	1.166e+00
379	1.237e-01
382	2.399e-01
383	1.790e-01
385	1.356e-01
386	4.845e-02
387	3.929e-01
388	1.474e-01
391	3.141e-02
392	5.457e-01
395	8.032e-01
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Coordenada y	Valor y (Dual)
396	5.342e-01
397	1.111e-01
398	4.545e-01
399	1.366e-01
400	8.369e-01
401	2.741e-02
402	5.945e-02
403	2.972e-01
409	9.369e-01
411	3.425e-01
413	8.318e-02
415	5.803e+00